Inversion of the z-Transform: Partial Fraction Expansion

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Key Concepts

1) The partial fraction expansion inversion method is for rational z-transforms. The basic idea is to decompose $X(z)$ into a sum of terms that can be inverted by inspection.

2) The procedure presented here assumes $X(z)$ is the ratio of an Mth-order polynomial in z^{-1} to an Nth-order polynomial in z^{-1}.

    a) If $M \geq N$, use long division to write $X(z)$ as an $M-N$-order polynomial in z^{-1} plus a proper rational polynomial.

    b) Factor the denominator of the proper rational polynomial as a sum of first-order terms involving the poles.

c) Invert each first-order term using the region of convergence and the known z-transform of exponential sequences.

d) Poles that occur in complex-conjugate pairs must have partial fraction expansion coefficients that are complex conjugates if $X(z)$ has real coefficients.

e) Information besides the region of convergence may be used to invert each first-order term.

    i) If the signal is known to be causal (right sided), then the right-sided inverse z-transform is chosen.

    ii) if the signal is known to be stable (absolutely summable), then the region of convergence includes the unit circle.

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Video Screenshots

Partial Fraction Expansion – inversion of rational functions of z^{-1}

Break $X(z)$ into elementary forms that can be inverted by inspection

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{8}{1 - z^{-1}} - \frac{9}{1 - \frac{1}{2}z^{-1}} + 2$$

$|z| > 1$

easy to invert

"undo" process of combining with a common denominator

$$X(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

write in powers of z^{-1}

1) If $M \geq N$ use long division to write

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \frac{\sum_{k=0}^{N-1} b'_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

2) Factor denominator as product of 1st order terms

$$\frac{\sum_{k=0}^{N-1} b'_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} = \frac{\sum_{k=0}^{N-1} b'_k z^{-k}}{a_0 \prod_{k=1}^{N} (1 - d_k z^{-1})}$$

3) Use partial fraction expansion in first-order terms

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$$\frac{\sum_{k=0}^{N-1} b'_k z^{-k}}{a_0 \prod_{k=1}^{N} (1 - d_k z^{-1})} = \sum_{k=1}^{N} \frac{A_k}{1 - d_k z^{-1}} \quad \text{(assuming distinct } d_k\text{)}$$

if d_i repeated r times $\quad \sum_{\ell=1}^{r} \frac{A_i^\ell}{(1 - d_i z^{-1})^\ell}$

4) Invert each term using ROC

$$\frac{A_k}{1 - d_k z^{-1}} \quad \overset{z}{\longleftrightarrow} \quad \begin{cases} A_k d_k^n u[n] & \text{for } |z| > |d_k| \\ -A_k d_k^n u[-n-1] & \text{for } |z| < |d_k| \end{cases}$$

$$\frac{A}{(1 - d_k z^{-1})^\ell} \quad \overset{z}{\longleftrightarrow} \quad \begin{cases} A \frac{(n+1)\cdots(n+\ell-1)}{(\ell-1)!} d_k^n u[n] & \text{for } |z| > |d_k| \\ -A \frac{(n+1)\cdots(n+\ell-1)}{(\ell-1)!} d_k^n u[-n-1] & \text{for } |z| < |d_k| \end{cases}$$

Example: $X(z) = \dfrac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})(1 - z^{-1})}$ ROC $1 < |z| < 2$

1) $M = 2 < N = 3 \Rightarrow$ no long division

2) Denominator is Factored

3) Part Fraction Expansion

$$X(z) = \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 - 2z^{-1}} + \frac{A_3}{1 - z^{-1}}$$

$$A_1 = X(z)(1 - \tfrac{1}{2}z^{-1})\Big|_{z^{-1} = 2} = \frac{1 - z^{-1} + z^{-2}}{(1 - 2z^{-1})(1 - z^{-1})}\Big|_{z^{-1} = 2} = \frac{1 - 2 + 4}{(1 - 4)(1 - 2)} = \frac{3}{3} = 1$$

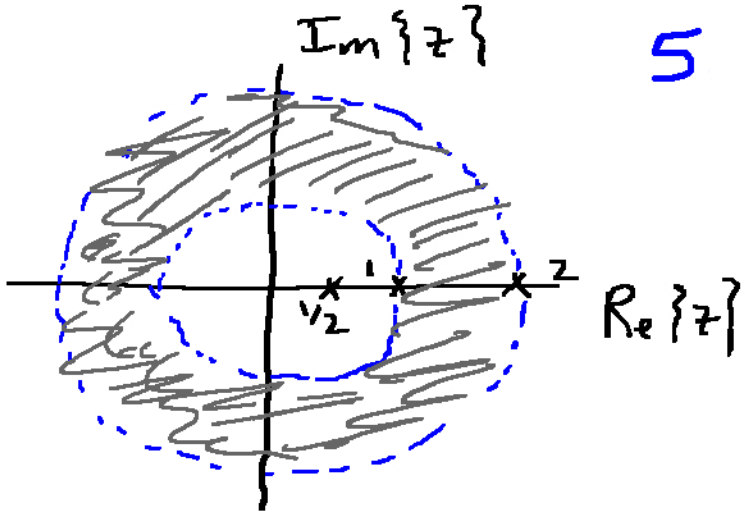
$$A_2 = X(z)(1 - 2z^{-1})\Big|_{z^{-1} = 1/2} = 2$$

$$A_3 = X(z)(1 - z^{-1})\Big|_{z^{-1} = 1} = -2$$

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{2}{1 - 2z^{-1}} - \frac{2}{1 - z^{-1}}$$

4) Invert each term using ROC info

ROC: $1 < |z| < 2$



$$\frac{1}{1-\frac{1}{2}z^{-1}} \quad \overset{z}{\longleftrightarrow} \quad \left(\frac{1}{2}\right)^n u[n]$$

$$\frac{2}{1-2z^{-1}} \quad \overset{z}{\longleftrightarrow} \quad -2\,(2)^n u[-n-1]$$

$$\frac{-2}{1-z^{-1}} \quad \overset{z}{\longleftrightarrow} \quad -2\,u[n]$$

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \quad -2\,(2)^n u[-n-1] \quad -2\,u[n]$$

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Example: $X(z) = \dfrac{z^3 - 10z^2 - 4z + 4}{2z^3 - 2z^2 - 4z} \qquad 0 < |z| < 1$

0) Write in powers of z^{-1}

$$X(z) = \frac{1 - 10z^{-1} - 4z^{-2} + 4z^{-3}}{2 - 2z^{-1} - 4z^{-2}}$$

1) $M = 3 > N = 2$ use long division

$$\begin{array}{r|l} & -z^{-1} + 3/2 \\ \hline -4z^{-2} - 2z^{-1} + 2 & 4z^{-3} - 4z^{-2} - 10z^{-1} + 1 \\ & \underline{4z^{-3} + 2z^{-2} - 2z^{-1}} \\ & -6z^{-2} - 8z^{-1} + 1 \\ & \underline{-6z^{-2} - 3z^{-1} + 3} \\ & -5z^{-1} - 2 \end{array}$$

$$X(z) = -z^{-1} + 3/2 + \frac{-5z^{-1} - 2}{2 - 2z^{-1} - 4z^{-2}}$$

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2) Factor denominator

$$X(z) = -z^{-1} + 3/2 + \frac{-5z^{-1} - 2}{2(1+z^{-1})(1-2z^{-1})}$$

3) Partial fraction expansion

$$\frac{-5/2\, z^{-1} - 1}{(1+z^{-1})(1-2z^{-1})} = \frac{A_1}{1+z^{-1}} + \frac{A_2}{1-2z^{-1}}$$

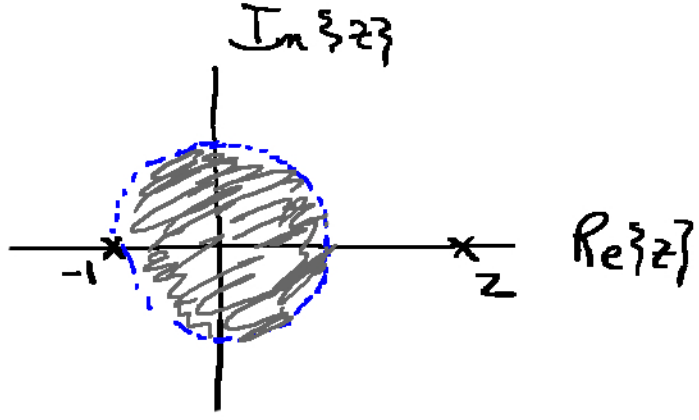
$$A_1 = \left.\frac{-\frac{5}{2}z^{-1} - 1}{1-2z^{-1}}\right|_{z^{-1}=-1} = 1/2$$

$$A_2 = \left.\frac{-\frac{5}{2}z^{-1} - 1}{1+z^{-1}}\right|_{z^{-1}=1/2} = -3/2$$

$$X(z) = -z^{-1} + 3/2 + \frac{1/2}{1+z^{-1}} - \frac{3/2}{1-2z^{-1}}, \quad |z| < 1$$

4) Invert terms using ROC

$$x[n] = -\delta[n-1] + 3/2\,\delta[n] - \frac{1}{2}(-1)^n u[-n-1] + \frac{3}{2}(2)^n u[-n-1]$$



Remarks –

1) Poles of $X(z)$ with real coefficients occur in complex conjugate pairs: $d_1 = d_2^*$

PFE coefficients must be in complex conjugate pairs

$$\frac{A_1}{1-d_1 z^{-1}} + \frac{A_1^*}{1-d_2 z^{-1}} = \frac{A_1^R + jA_1^I}{1-d_1 z^{-1}} + \frac{A_1^R - jA_1^I}{1-d_2 z^{-1}}$$

2) Information other than ROC

- causal signal $\Rightarrow$ right-sided inverses
- stable signal $\Rightarrow$ ROC includes $|z|=1$

$$\frac{z}{z-d} \longleftrightarrow d^n u[n]$$

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